Applied AI in Biomedicine

The Improvised Radiologists

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1 Introduction

The goal of this project is to develop and train deep learning models for the **classification** and **segmentation** of **breast lesions** in medical imaging, within the context of breast cancer diagnosis. Based on different modeling strategies and the use of masked images, two distinct pipelines were implemented and their performances systematically compared.

2 Materials and Methods

2.1 Dataset

The dataset consists of *1531 ultrasound images* acquired from *968 patients*, with multiple images potentially corresponding to the same patient. Each image is associated with a binary mask delineating the precise lesion region.
For the classification task, three classes are defined: **Normal (N)** images with no region of interest (ROI), **Benign (B)** lesions characterized by smooth boundaries, and **Malignant (M)** lesions exhibiting irregular, jagged shapes.

2.2 Outlier Detection

Prior to dataset splitting, an outlier analysis was performed to verify the consistency between the binary masks and their corresponding class labels. The analysis confirmed the coherence between lesion masks and class labels.

2.3 Data Splitting

Prior to dataset splitting, a majority label was assigned to each patient based on the most frequent class among their associated images. This patient-level label was used to preserve the original class distribution while enforcing a patient-wise split, thereby preventing information leakage arising from correlated samples belonging to the same subject.

To evaluate the final model performance, the dataset was divided into two cohorts (**80% - 20%**): one for **training and validation** and one for **internal testing**. Following this split, the effective training set comprised *1195 images*.
For the classification task, **5-Fold Stratified Cross Validation** was employed on the training cohort to ensure balanced class representation across folds while maintaining patient-level separation. For the segmentation task, the data were instead randomly divided into two subsets to reduce computational overhead.

2.4 Pre-Processing

Several pre-processing techniques were applied to the images to improve contrast enhancement and ensure intensity normalization. In order:

- **Resizing**: since the original images varied in size, they were first converted to square format by adding padding and subsequently resized to match the input resolution required by each model (256×256 px, 300×300 px, and 512×512 px).
- **Crop & Zoom** (only for Task 2B): to perform a lesion-aware crop of the input images thanks to the binary masks.
- **Inpainting**: to remove annotations and imaging artifacts that were present in a subset of the images, thereby preventing potential bias and residual visual cues that could inadvertently influence the learning process. This step was particularly important given that the dataset was collected from multiple institutions, where differences in annotation protocols, acquisition settings, and post-processing practices resulted in heterogeneous artifact patterns. By applying inpainting, we aimed to standardize the visual appearance of the images across organizations and ensure that the model learned clinically relevant features rather than institution-specific or

annotation-related artifacts. Some examples are visible in Figure 1.

- **Noise Filtering**: a median filter was implemented to remove typical ultrasound noise, such as speckle noise. Other noise-reduction techniques reported in the literature (including a bilateral filter combined with a Wiener filter and wavelet-based methods) were also evaluated; however, the median filter consistently provided the best results.
- **Normalization**: Contrast enhancement was performed using CLAHE, a technique well suited for ultrasound imaging[1], followed by Z-score normalization to standardize image intensity distributions.

Some examples of complete pre-processing pipeline is reported in Figure 2.

2.5 Data Augmentation

Given the reduced size of the dataset, we implemented some data augmentation techniques applied on-the-fly by the data loaders to increase data variability and to strengthen the generalization capability of the models. Geometric transformations were used coordinately on image and mask, while intensity transformations were applied exclusively on images.
The geometric data augmentations applied included:

- **Horizontal and Vertical Flips**
- **Random Rotation**
- **Random Affine**
- **Elastic Deformation**

Photometric augmentations were also explored, including:

- **Brightness & Contrast**
- **Coarse Dropout**

All augmentation parameters were selected based on established practices in the ultrasound imaging literature for lesion recognition and further refined through visual inspection to ensure anatomical plausibility. No class-specific augmentation strategies were employed, as class imbalance was addressed through the choice of loss functions.

Some examples of data augmentation can be seen in Figure 3.

2.6 Loss Functions and Metrics

Several loss functions and evaluation metrics were explored to address class imbalance in the classification task and to improve boundary delineation in the segmentation task.

- **Weighted dice score**
- **Weighted or not weighted CE**
- **Boundary loss functions**
- **Focal loss**
- **Lovász-Softmax**[2]
- **Tversky Loss**[3]

Model performance was evaluated using multiple metrics, as reported in Table 1. We highlight the importance of **Recall** in detecting false negatives, which is especially important in the context of this challenge.

METRIC	TASK	NOTES
Balanced Accuracy	C	Main
Dice	S	Main
Hausdorff Distance	S	Boundaries
Recall	C	FN
Precision	C	FP
IoU	S	Main

Table 1: Metrics

2.7 Task 1A - 3-Classes Classification

The goal of this task is to train a model able to classify images into three classes, using numerical labels: 0 for Benign, 1 for Malignant, 2 for Normal.

Since the masks represent an inestimable information to detect the ROIs, and patches of the images cropped around the lesion areas with some context are more easily identifiable with respect to the the full images, we first opted to train a **dual-stream classifier architecture** that would receive for the first stream the mammography images, while for the second stream the associated masks. By adopting **ResNet50** pretrained on ImageNet for the first stream and **ResNet18** trained from scratch for the masks, a balanced **validation accuracy over 0.90** was obtained. We also trained **EfficientNetB3**[4] for the first stream and obtained better and more stable results for the classification.

However, since lesion masks are not available at test time and they would have introduced bias, we explored alternative approaches that do not rely on explicit mask information. To build upon our previous experiments, we introduced a **localizer** (an **attention U-Net** trained with a weighted loss—to automatically identify lesion regions). The predicted lesion area was then used to crop the image (a central crop was applied to Normal images), preserving a limited amount of surrounding context. The resulting crops were subsequently provided as input to the dual-stream classification model based on EfficientNet-B3.

To mitigate the destructive propagation of errors, we implemented a dual-stream classifier consisting of a **global** and a **local stream**. The global stream received the full image to capture contextual information, while the local stream processed a cropped region—either around a lesion identified by the same localizer as before or a central crop for non-lesion cases. For the global stream, we used a **ResNet18**, and for the local stream, an **EfficientNet-B3** model. Both streams were trained using a **knowledge distillation paradigm**: the global stream distilled knowledge from a **ResNet50** trained on full images, and the local stream from the dual-stream **EfficientNet-B3**. The outputs of the two streams were then merged to produce the final classification. Despite implementing recall maximization and fallback strategies, this architecture remained highly sensitive to error propagation, limiting its overall performance.

From these experiments, we observed that ResNet excels at capturing global context, while EfficientNet is better at distinguishing fine-grained details. To leverage both strengths, we opted for a **ResNeXt** model, which extends the classical ResNet architecture by integrating multi-path learning and an attention mechanism. In ResNeXt, the feature map of an image is split into cardinality groups, and each group is further divided into subgroups that undergo independent convolutions. An attention weight is computed for each group, and the most relevant features are highlighted via a weighted sum. This mechanism is robust to noise, adapts to local contrast, and allows the model to learn complex spatial correlations among channels—critical to distinguish Malignant from Benign lesions.

2.8 Task 1B - B and M Segmentation

The goal of this task is to train a model capable of accurately segmenting images that were previously classified as M or B (containing a lesion). Binary segmentation masks were used as ground truth annotations, and model performance was quantitatively assessed using the Dice coefficient.
Initially, we considered adopting **MONAI (Medical Open Network for AI)**[5], an open-source domain-specific framework, specifically designed to support deep learning applications in medical imaging with a strong emphasis on segmentation. MONAI provides a rich ecosystem of components, including highly customizable pre-processing and data augmentation transforms tailored to medical images, a variety of pre-defined network architectures commonly used in clinical applications, and medical imaging–specific loss functions and evaluation metrics.Despite the ease of integration and the maturity of the framework, several MONAI modules exhibited limited transparency, effectively behaving as black boxes. This reduced our ability to perform fine-grained architectural modifications, experiment with non-standard training strategies, or introduce custom inductive biases tailored to the characteristics of our dataset.

To establish strong baselines, we experimented with multiple MONAI-provided architectures, including **U-Net**, **Attention U-Net**, and **DynUNet**. While these models demonstrated reliable convergence and reasonable segmentation performance, they consistently underperformed compared to our custom-designed architectures, namely **ResFormer** and **SegFormer**[6]. The performance gap became particularly evident in challenging cases involving low-contrast lesions, irregular boundaries, or anatomically complex regions. Both ResFormer and SegFormer incorporate **transformer-based self-attention mechanisms**, which allow explicit modeling of long-range spatial dependencies and global anatomical context[7]. This capability is especially critical in medical imaging, where accurate lesion delineation often depends not only on local texture cues but also on contextual information from surrounding tissues and distant anatomical structures. By contrast, purely convolutional architectures rely on progressively enlarged receptive fields and are inherently limited in capturing non-local relationships[8]. Although ResFormer benefited from the integration of residual learning with transformer blocks, **SegFormer** ultimately achieved superior segmentation performance. This improvement can be attributed to SegFormer's hierarchical **Mix Transformer (MiT) encoder**, which extracts **multi-scale features** while preserving spatial coherence, and its lightweight **MLP-based decoder**, which efficiently fuses features from

different scales without introducing excessive parameters[9]. This architectural design results in more effective multi-scale representation, improved training stability, and stronger generalization capabilities – advantages that are particularly important in limited-data medical imaging scenarios, where overfitting remains a major concern. Overall, these results indicate that **transformer-based architectures**, and **SegFormer** in particular, offer a more suitable balance between global context modeling and spatial precision for medical image segmentation compared to traditional **U-Net**–based approaches.

The model has been trained using a **2-fold cross validation** and a low learning rate, dynamically driven by **Ranger21**, including the warm up. As **SegFormer** suffers from small batch size for convergence stabilization and for this task it is crucial to have high resolution (512x512), increasing the probability of facing the *Out-Of-Memory* error, we used **Gradient Accumulation** to simulate a larger batch size. **DiceCELoss** was used to improve training stability. Some boundary losses (Boundary Loss from **MONAI** and **HausdorffDTLoss**) were considered due to their focus on lesion boundaries, an especially important feature in malignant lesions, but revealed to be a major cause of training instability.

To balance model complexity and the limited dataset, a strong on-the-fly **data augmentation**, comprising Random Affine transformations, was implemented. Furthermore, both **Early Stopping** and a strong **weight decay** were used. Finally, some **post-processing** was used to sharpen boundaries through binarization, morphological operations (opening and closing to remove artifacts) and *GrabCut*.

To emphasize the borders of lesions and accurately capture their shapes, we employed a **combined loss function** for this task, consisting of: **Dice Loss**, a standard baseline for segmentation tasks; **Lovász-Softmax**[2], which optimizes the **IoU** metric and generally provides better convergence than Cross Entropy; **Tversky Loss**[3], which balances precision and recall to control false positives and false negatives; **Boundary Loss**[12], which computes the distance from the ground truth borders to guide precise boundary prediction; **Edge-Aware Focal Loss**[13], a variant of Focal Loss that focuses on morphological edge pixels; **Contour Loss**, a variant of Binary Cross Entropy Loss designed for the contour-aware decoder; **Empty Penalty**, a custom penalty term that discourages false positive predictions in empty regions.

During training we adopted different strategies to improve model performance: **Detection Gating**, meaning the model is allowed to learn both segmentation and lesion identification freely during the first 10 epochs (**Gating Warmup**). Afterward, if the identification confidence falls below a threshold (0.6), the output segmentation mask is forced to be empty (black); **Cosine Annealing LR**, this schedule provides smooth convergence and helps prevent sudden drops in performance; **Gradient Clipping**, applied to avoid exploding gradients, ensuring stable training; **Early Stopping** (setting the patience at 10 epochs) so that the best model was actually saved only when the validation error was lower than the training error.

2.9 Task 2A - 3-Classes Segmentation

The goal of this task is to train a model for image segmentation. Unlike the previous classification step, the model must first determine whether a lesion is present in the image. If a lesion is detected, the model should segment it and generate a corresponding binary mask; if no lesion is present, the output mask should remain empty, showing no region of interest (ROI).

For this step, we used **Segformer**[6], a transformer-based autoencoder, enriched with: a **detection head**, in order to determine whether the image contains a lesion; a **contour-aware decoder**[10], which aims to supervise the boundary prediction predicting edge maps after fusing multi-scale features; **deep supervision**[11], providing additional classification heads at intermediate encoder stages to encourage early layers to learn discriminative features.

2.10 Task 2B - B vs M Classification

The goal of this task is to train a model to classify masked images into one of the two lesion classes. The masked images are generated using the masks predicted in the previous segmentation task, removing background so that only the lesion is visible. During training, the provided binary masks were used to create the masked images after pre-processing. The mask boundaries were slightly expanded to simulate a pessimistic segmentation scenario.

For this step, we used **EfficientNet-B3**[4], a CNN backbone with compound scale, enriched with: **ROI-focus Attention**, so that the model could totally ignore the black background; an **Edge-Aware mod-**

ule, which extracts edge irregularity features through **Sobel**[14] gradient, so it can discriminate the lesion according to its shape; **Feature fusion**, in order to take into consideration also the lesion texture in discrimination operation; an **auxiliary head**, to support the classification of lesion borders and to learn morphological patterns.

During training we combined several loss functions, such as **Weighted Focal Loss**, to handle class imbalance and **Auxiliary Edge Loss**, as a support loss for the Edge-Aware module. **Label smoothing** was used to prevent overconfident predictions.

Other training strategies were: a **high dropout** (0.6) to provide stronger regularization given the limited dataset size; **up-weighting** of the minority class to address imbalance; **Cosine Annealing LR**, for smooth convergence; **Early Stopping** (setting the patience at 10 epochs), saving the best model only when it did not overfit on the validation set.

3 Results

Here are reported the training and testing results for each sub-task. Additional results are analyzed using confusion matrices and performance plots in the Appendix.

Task 1A As previously reported, **ResNeSt-50** was selected as the final model for this task. It was trained using 5-fold cross-validation, weighted cross-entropy loss, and gradient clipping and accumulation to ensure stable training. To mitigate overfitting without reducing the available training data, strong weight decay and dropout were applied. The best model from each fold was selected based on balanced accuracy on the validation set. For testing, the models from all folds were **ensembled**, and **Test-Time Augmentation (TTA)** using the same training augmentations was applied. Performance is summarized in plots and confusion matrices in Figure 4.

	Validation	Testing
Fold 1	0.7912	
Fold 2	0.8415	
Fold 3	0.8210	0.8227
Fold 4	0.8459	
Fold 5	0.8169	

Table 2: Balanced Accuracy for Task 1A

Task 1B The **SegFormer**[6] was evaluated using 2-fold cross-validation. While boundary-aware losses were explored, they led to unstable training and were therefore excluded from the final setup. **Post-processing** and **TTA** were applied to refine predictions. Training-Validation plots are reported in Figure 6.

	Validation	Testing
Fold 1	0.7368	0.5494
Fold 2	0.7309	

Table 3: Dice Score for Task 1B

Task 2A The model trained for this step is the **SegFormer**[6] previously described. Due to its high computational demands arising from the enriched encoder-decoder architecture, only a single model was trained, and cross-validation was not performed. **Segmentation results** during testing can be seen in Figure 8, whilst the **training plots** are shown in Figure 7.

Validation	Testing
0.7721	0.3694

Table 4: Dice Score for Task 2A

Task 2B For this task, an **Efficient-B3**[4] model has been trained through a 5-fold Cross Validation, as described above. During testing, we inferred the results through an **ensemble** of the models from the validation folds, and we applied **TTA** using the same augmentations used during training. The entire pipeline to build the **masked images** can be seen in Figure 9, whilst **plots and confusion matrices** are shown in Figure 10.

	Validation	Testing
Fold 1	0.9203	
Fold 2	0.9192	0.7111 S Masks
Fold 3	0.9059	0.9279 GT Masks
Fold 4	0.9087	
Fold 5	0.8992	

Table 5: Balanced Accuracy for Task 2B

4 Discussion and conclusions

At first glance, **Task 1** clearly outperformed **Task 2** on both the validation and testing sets. The main reason for Task 2's poorer performance is the **segmentation** subtask, which acts as a critical **bottle-neck**: when the segmentation produces numerous false positives (e.g., normal images incorrectly marked as containing lesions) or inaccurately captures the lesion shapes, the downstream classifier receives flawed input and cannot perform as well as it did during validation. In other words, errors in segmentation propagate directly, limiting the overall effectiveness of the classification pipeline.
Indeed, one of the main challenges of this project concerned the segmentation task, since **multiple models** had to be explored before identifying those that achieved satisfactory performance. A significant difficulty was the tendency of the models to **over-smooth lesion shapes**, leading to the loss of important information associated with malignant regions, hence particular attention was devoted to the preservation of **lesion boundaries**. Another challenge involved the detection of lesions in normal images, primarily caused by **shadow artifacts**. Although the proposed **gating method** demonstrated promising results, the final model still exhibited false positives. We tried to use ground truth masks from the test set to create the masked images, and the resulting performance was significantly better than when using the predicted segmentation masks, achieving results (Table 5) comparable to those observed during validation. For this reason, we retain the segmentation model accountable for the reduced performance of the classification model.

Based on the validation results, classification performance is consistently higher when the model is trained on **zoomed-in lesion crops** (**2B**) compared to **full-context** images (**1A**). This observation is consistent with the initial experiments using the double-stream model with ground-truth masks in **Task 1A**: here the best performance was achieved when the image stream was fed lesion-aware cropped images. Focusing the input on the lesion region appears to emphasize clinically relevant **morphological features** while reducing background noise, thereby enabling the model to learn more discriminative representations for lesion classification.

An the end, future work will primarily focus on reducing the performance gap observed between **Task 1** and **Task 2** by improving the segmentation stage, which as stated above currently represents the main bottleneck of the pipeline. In particular, the adoption of **pre-trained models** on datasets more closely related to the target domain could provide more robust and discriminative feature representations, especially for lesion boundaries, and mitigate the over-smoothing effects observed. Moreover, instead of relying on full-image inputs, a **patch-based training strategy** could be explored to allow the models to focus on local lesion characteristics, reducing the influence of background artifacts and shadows that led to false positives in normal images. Such an approach may help **Task 2** better approximate the performance achieved when ground truth masks are used, narrowing the gap with **Task 1**. Finally, exploring joint or end-to-end optimization of segmentation and classification, as well as incorporating multi-scale contextual information, could further limit error propagation between subtasks and lead to more consistent performance across validation and test sets.

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Appendix

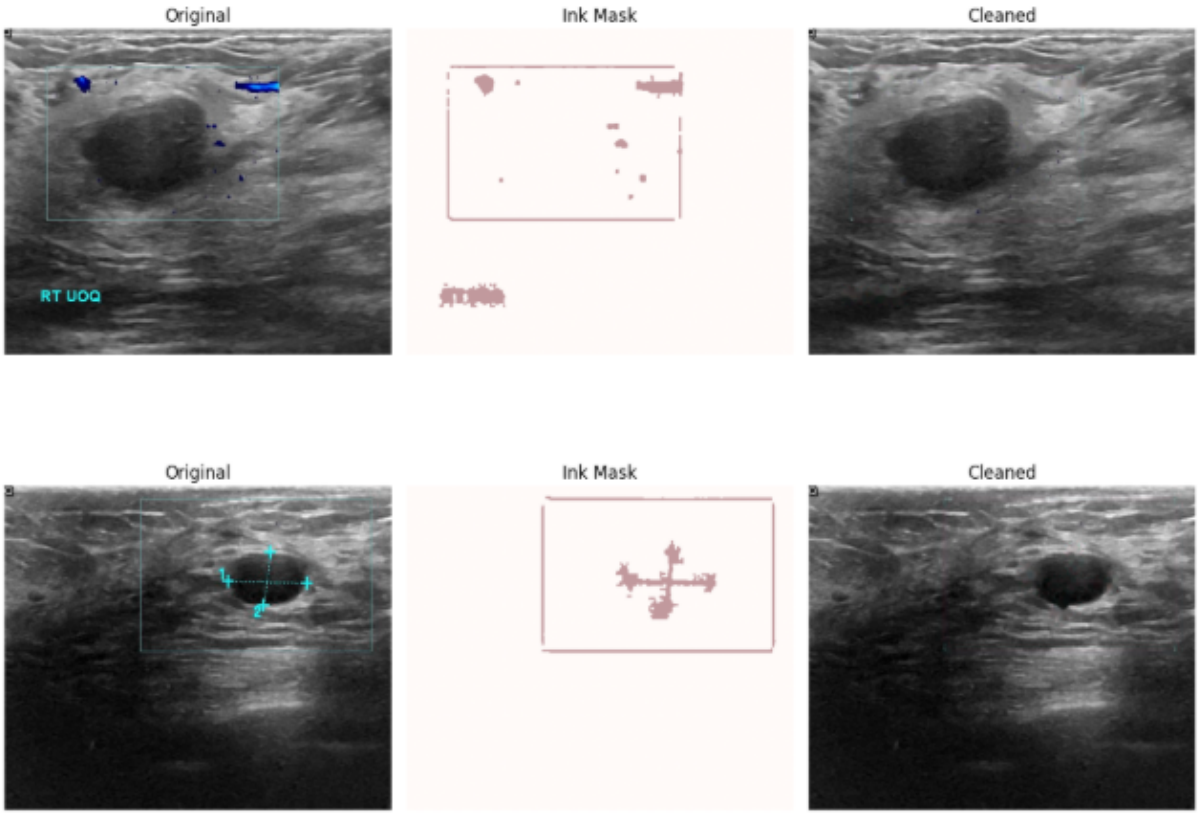


Figure 1: **Annotation detection**. a) Original image b) Annotation mask c) Cleaned image

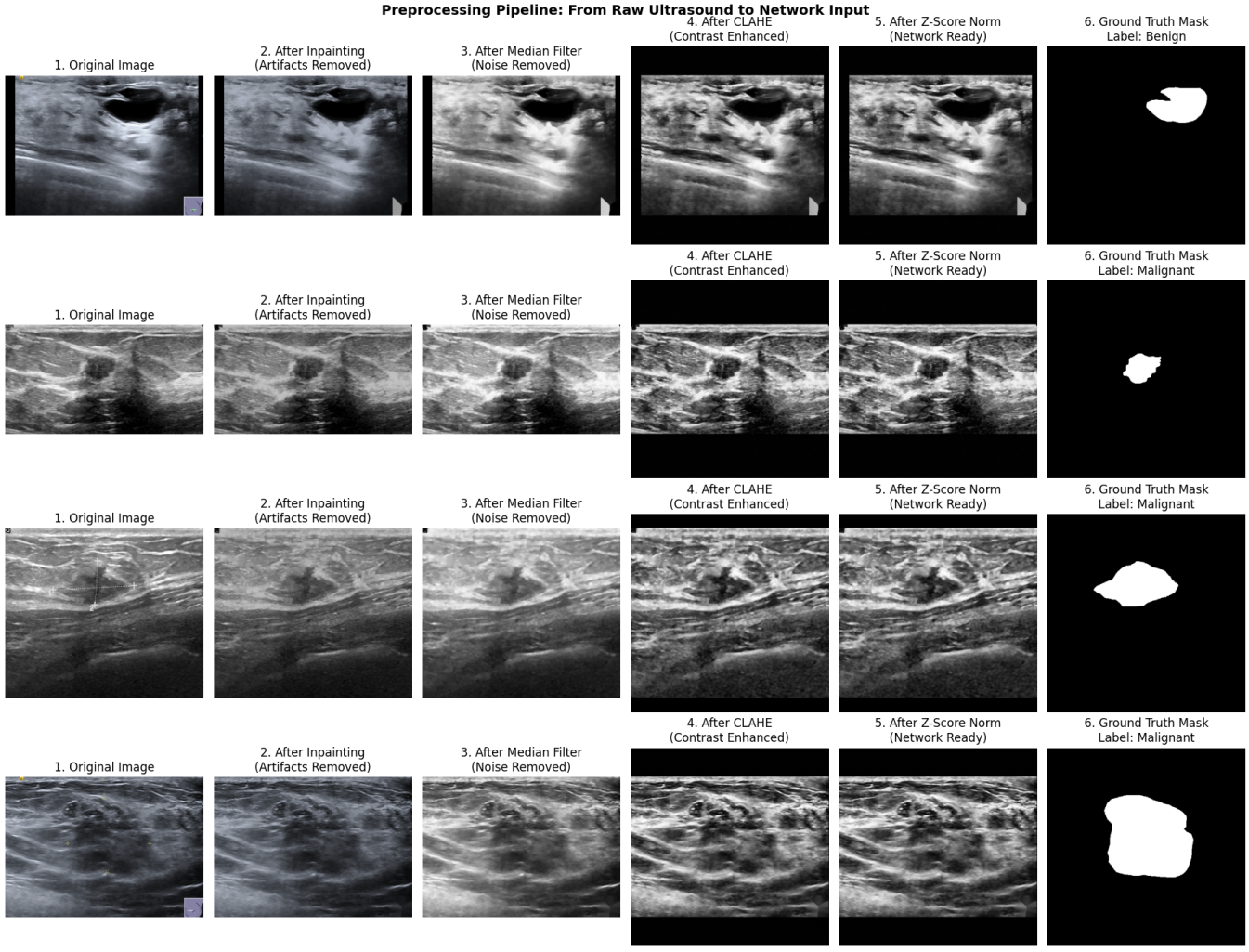


Figure 2: **Pre-processing pipeline**. a) Original image b) Inpainting c) Median filter d) CLAHE e) Z-Score Normalization f) GT Mask

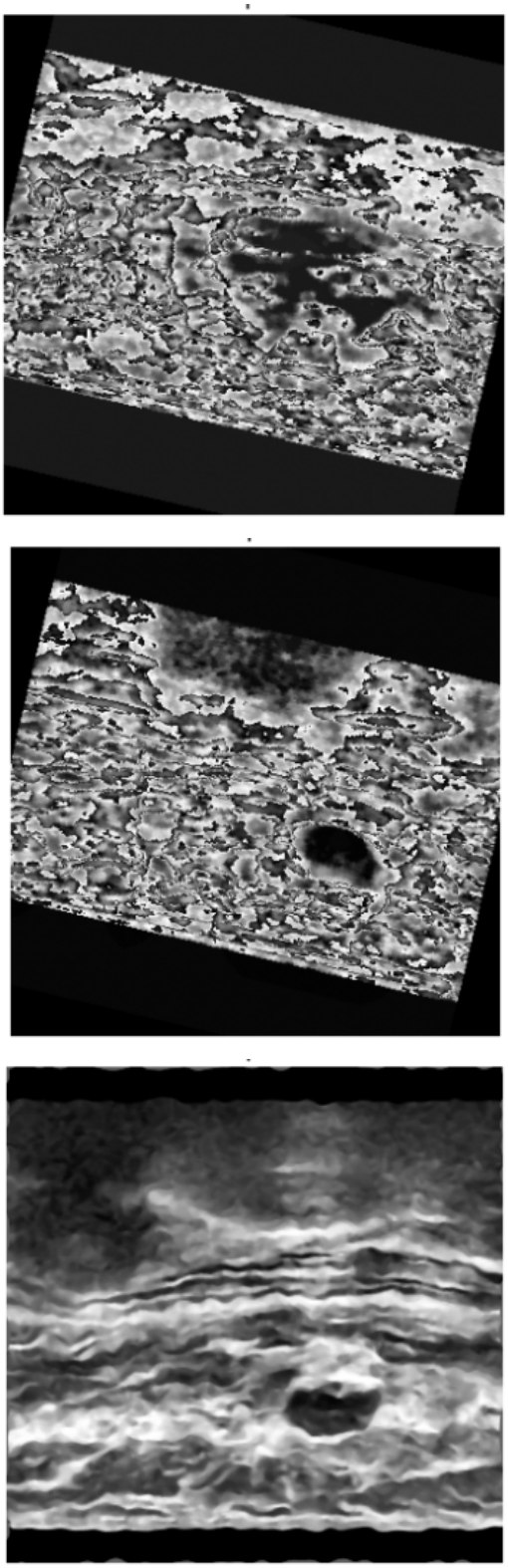


Figure 3: **Examples of data augmentation**. a) Brightness and rotation b) Elastic transform and rotation c) Elastic transform and vertical flip

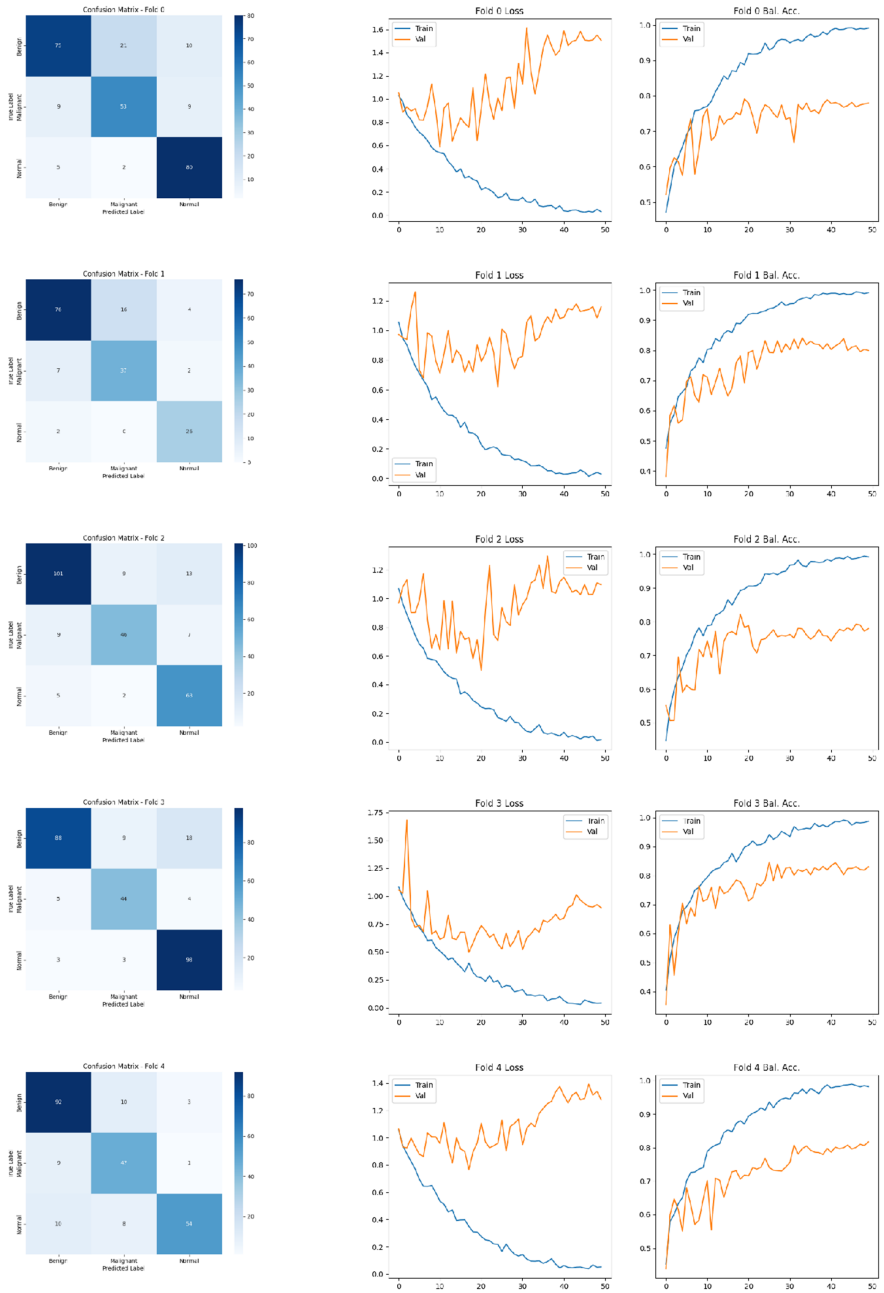


Figure 4: **Classification 1A - Training/Validation confusion matrices and plots**. On the left: the confusion matrices for each fold. On the right: Loss and BA plots. Note that early stopping hasn't been used for this training.

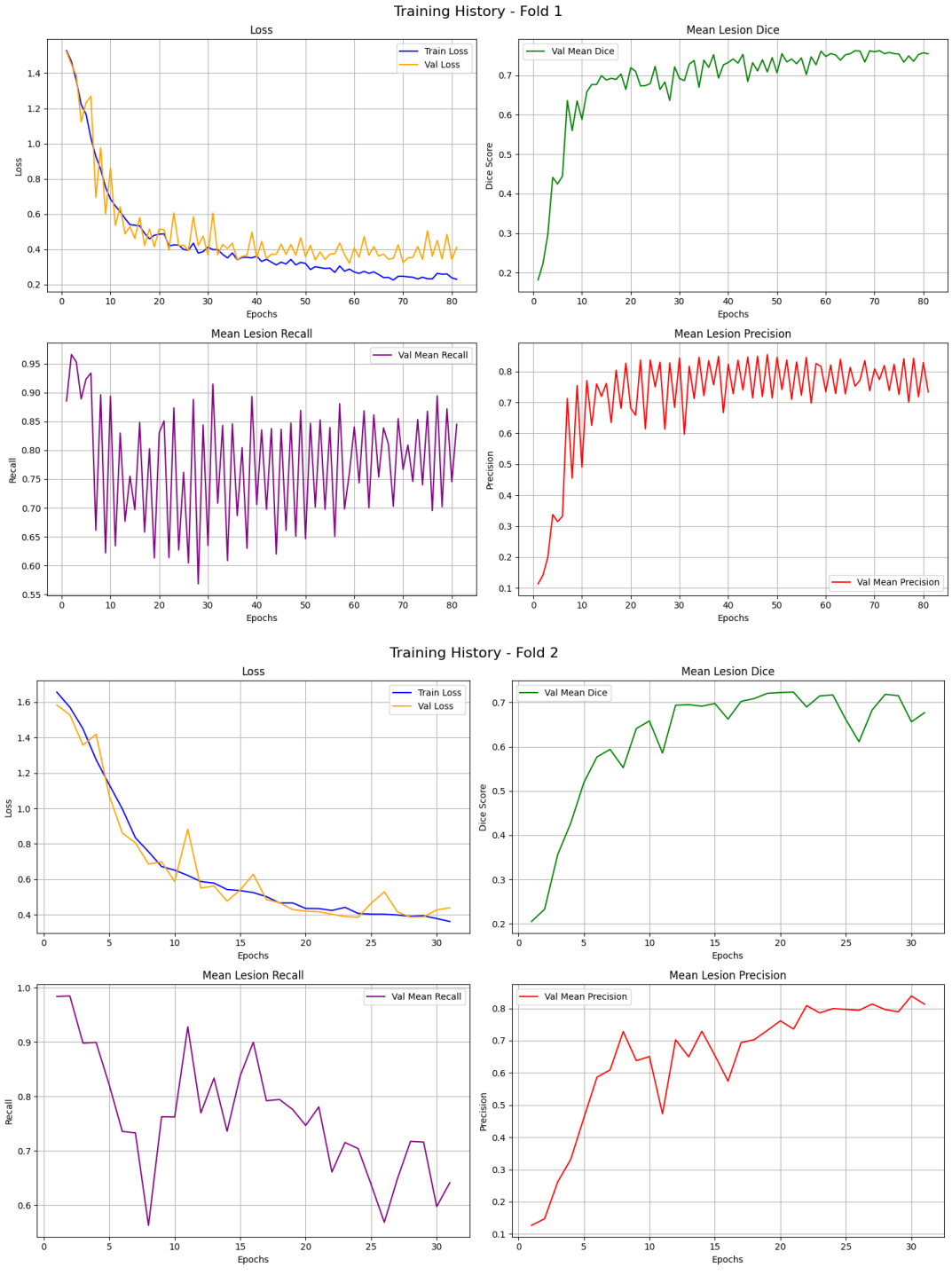


Figure 5: **Segmentation 1B - Training/Validation plots**. Loss, Dice score, Recall and Precision plots for each fold.

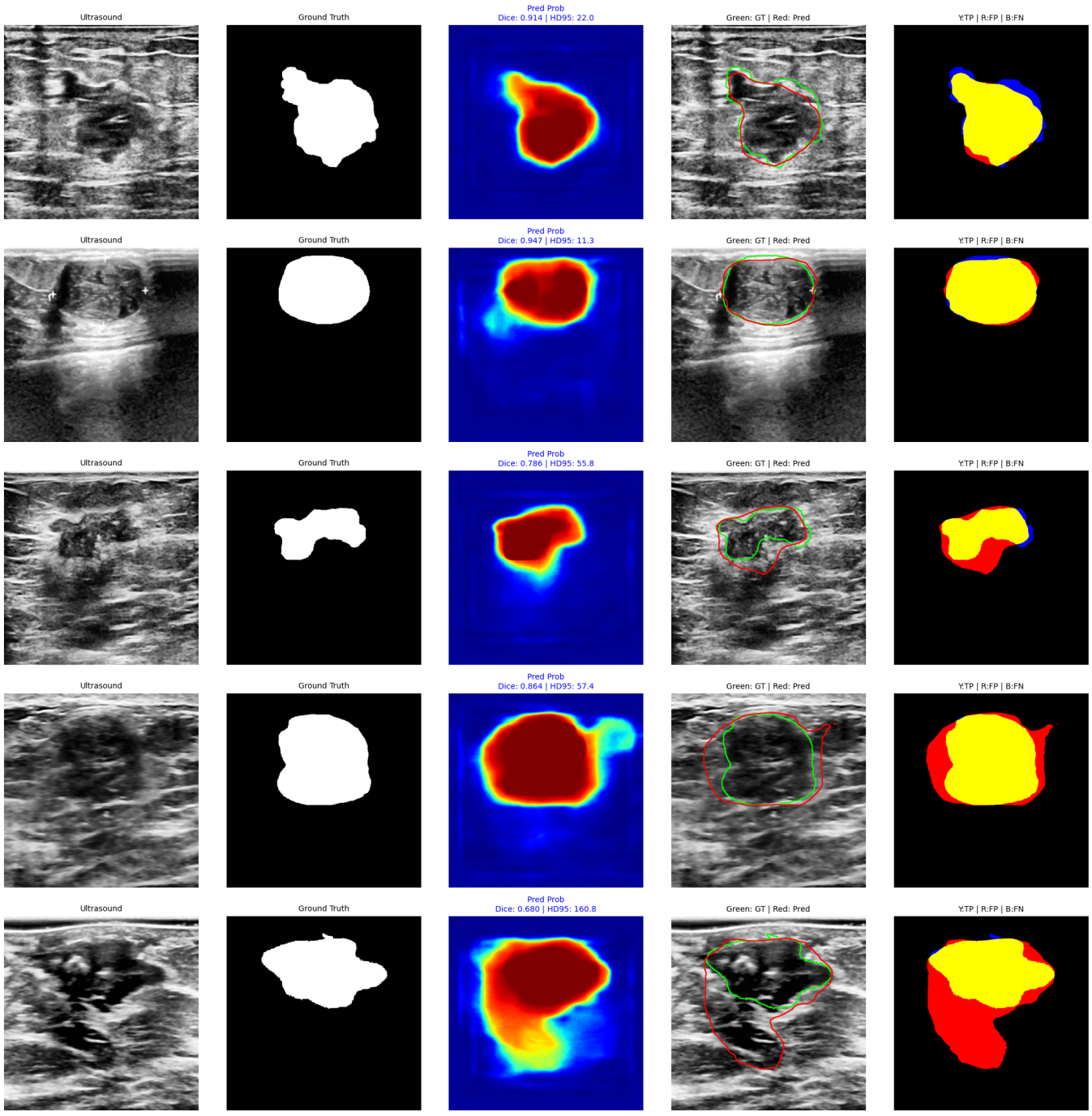


Figure 6: **Segmentation 1B - Visualization of results**. Green boundary is the ground truth. Red boundary the predicted one. Yellow area is true positive

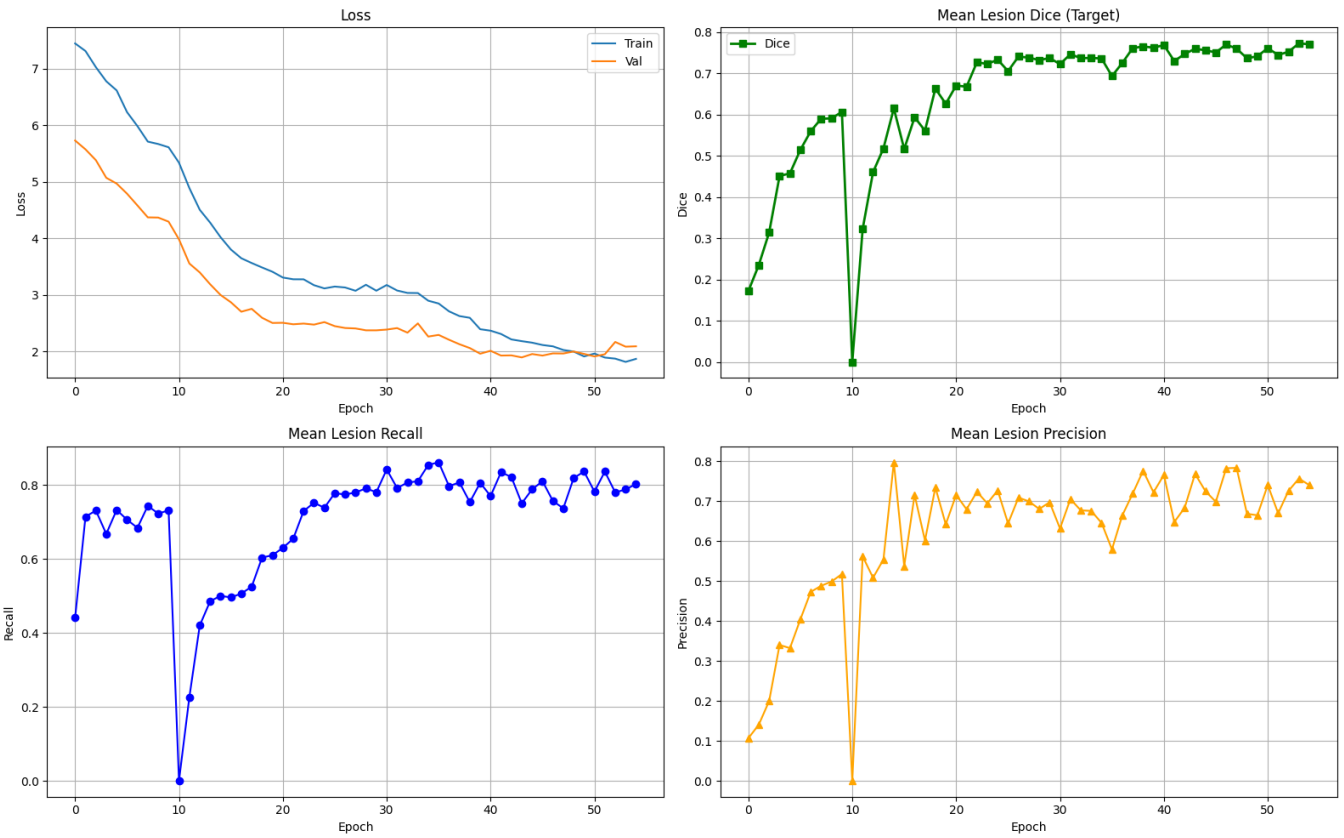


Figure 7: **Segmentation 2A - Training/Validation plots**. a) Training Loss in blue, Validation Loss in orange. b) Validation Dice Score in green. c) Validation Recall in blue. d) Validation Precision in yellow.

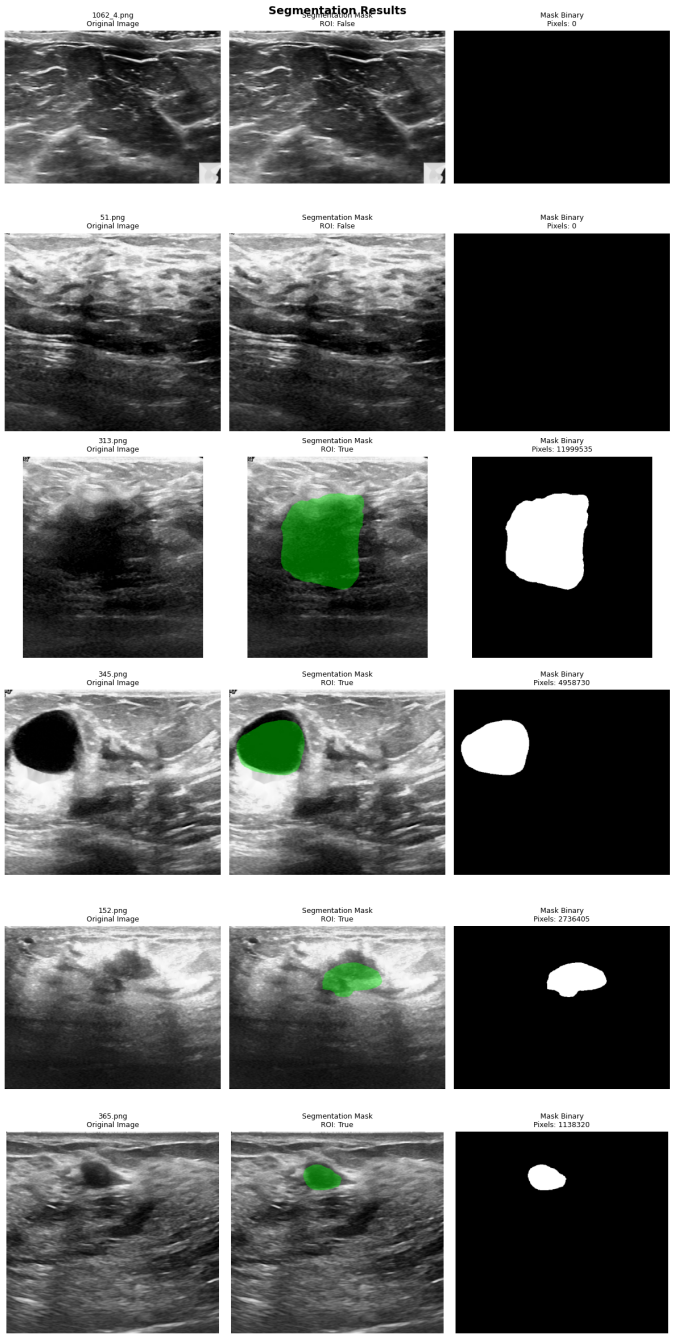


Figure 8: **Segmentation 2A - Results visualizations from testing set**. Six samples: first three are Normal images, the last three contain a B/M lesion.

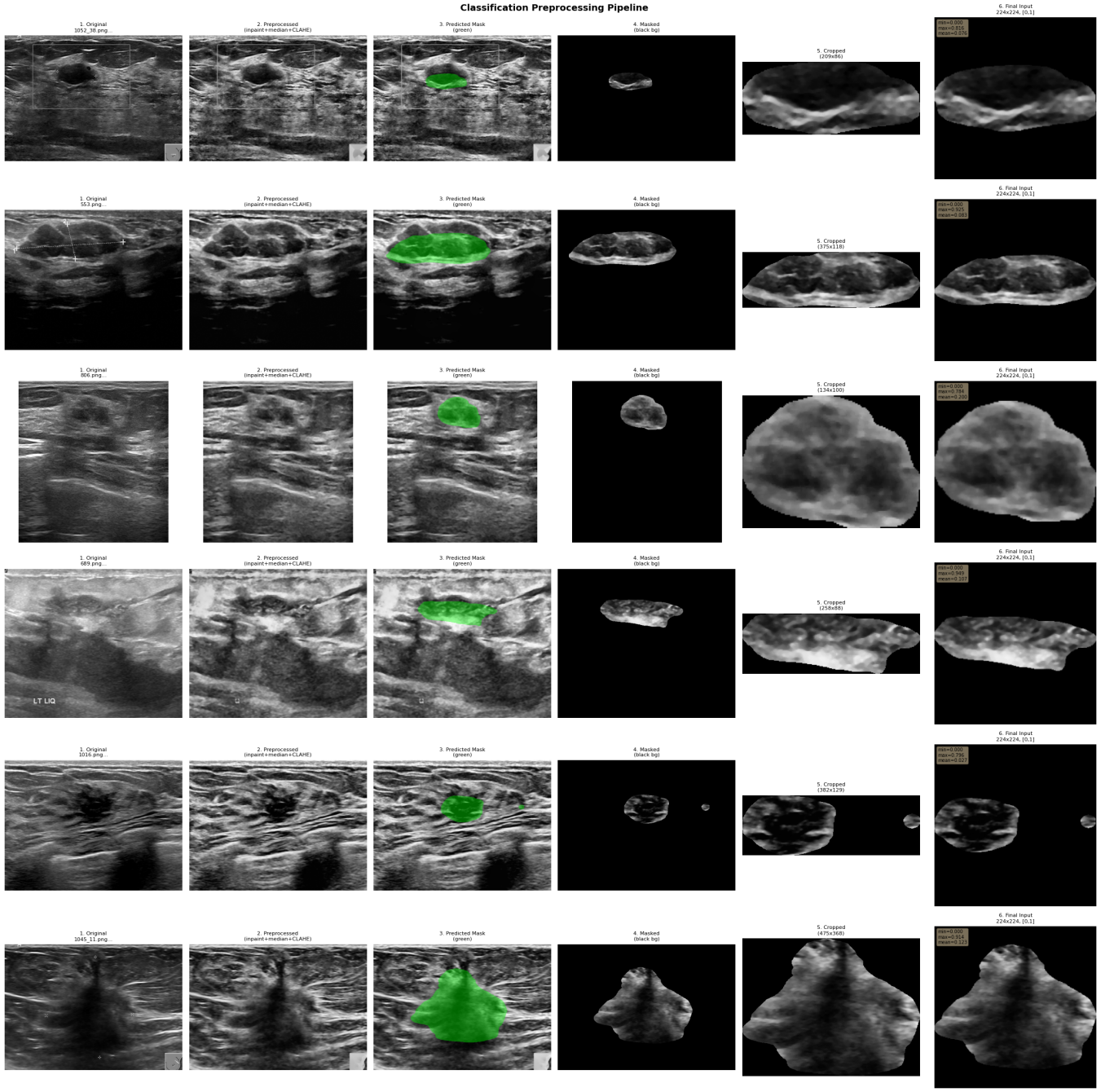


Figure 9: **Task 2 - Masked image building pipeline**. a) Original sample. b) Pre-processing. c) Segmentation prediction. d) Masked image using segmentation mask. e) Zoomed and cropped masked image ready for classification.

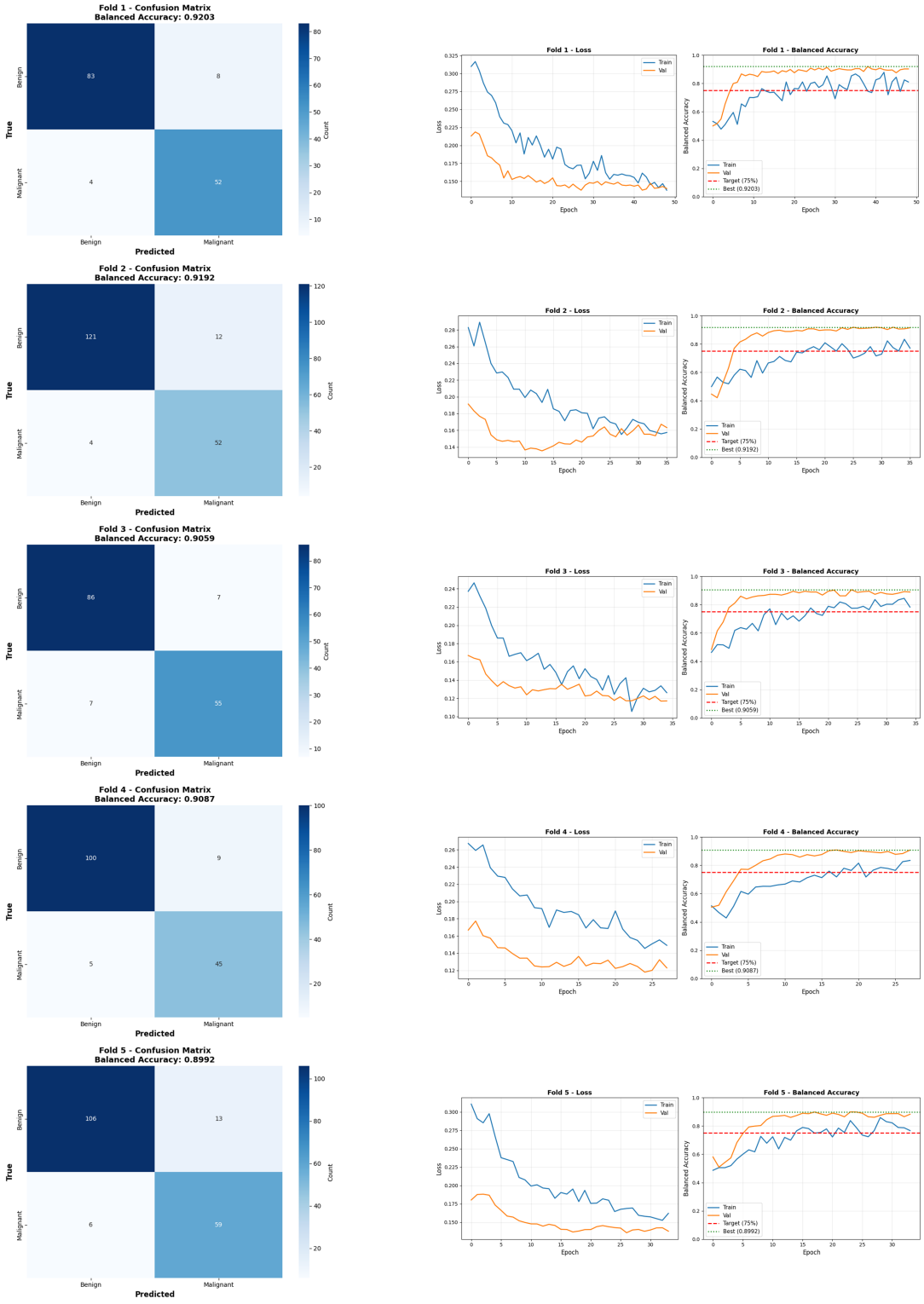


Figure 10: **Classification 2B - Training/Validation confusion matrices and plots**. On the left: the confusion matrices for each fold, on the right: the Loss and BA plots.